Teo Kai Jie (Kendrick)

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EDUCATION

Nanyang Technological University **Expected Jul 2028**
Bachelor of Computing (Honours) in Computer Science *Singapore*

- **Current GPA: 4.56/5.00** (Highest Distinction)

Singapore Polytechnic **Mar 2022**
Diploma in Information Technology *Singapore*

- **Final GPA: 3.80/4.00**

EXPERIENCE

Singapore Armed Forces **Dec 2022 – Aug 2024, May 2025 – Jul 2025**
Software Engineer (NSF) *Singapore*

- Developed a **Node.js/Express.js** and **EJS** email-based QR code attendance system to automate check-in for **300+ participants** in real time, earning a feature on Army News.
- Conducted functional **QA testing** across Python-based Telegram chatbots, isolating edge cases and resolving message-handling defects to stabilize the service for daily operational use.
- Designed functional components for a Microsoft PowerApps manpower tasking application and devised a mobile variant, resolving **300+ operational cases** after user testing.

EcoExchange Pte. Ltd. **Oct 2021 – May 2022**
Lead Frontend Developer (Industry Client Project) *Singapore*

- **Led a 3-man frontend team** to build a corporate intranet using **HTML5, CSS3, and JavaScript**.
- Digitized legacy paper-based operations into an interactive tracking portal, successfully processing **1,000+ live operational tickets**.

Leaptron Engineering Pte. Ltd. **Feb 2021 – Aug 2021**
Software Engineer (Intern) *Singapore*

- Built a QR scanner using **Android SDK (Java)** to replace manual data entry for warehouse personnel.
- Architected a **Node.js/MariaDB** warehouse management system from scratch, designing normalized schemas and testing **MySQL** stored procedures to process inventory transactions and reduce query execution times.

PROJECTS

Personal Website | *Website* | *GitHub* **May 2026 - Aug 2026**

- Built custom **TypeScript/Tailwind CSS** components in **Next.js**, versioned in **Git** and hosted on **Vercel**.
- Developed an **LLM chatbot** using **Gemini API** and **ElevenLabs API** to stream real-time voice responses synthesized from a custom voice model.
- Devised a **WebGL** 3D voxel viewer in **Three.js**, using **InstancedMesh** batching and Intersection Observers to pause off-screen rendering and maintain **~60 FPS**.

CellCollab | *Website* | *GitHub* **May 2025 - Oct 2025**

- Created a real-time multiplayer cellular automata playground using **React.js** and deployed via **Heroku**.
- Leveraged **Socket.IO** to enable **<100 ms** synchronization for live cell edits and active brush hover positions.
- Designed a **Node.js** simulation engine that executes state calculations at **100 ms/tick** and broadcasts board updates across SHA-256 hashed rooms.
- Wrote a **Python** data script to scrape and convert Conway's Life Lexicon pattern assets into 2D binary arrays for brush presets.

SKILLS

- **Languages**: TypeScript, JavaScript, HTML5, CSS3, Java, Python
- **Frameworks:** Node.js, Express.js, Socket.IO, React.js, Next.js, Android SDK (Java), Tailwind CSS, Three.js
- **Databases:** MySQL, MariaDB
- **Developer Tools & Deployment:** Git, GitHub, Vercel, Heroku